

PROJECT SUMMARY

[GitHub Repository](#)

Customer Churn and Revenue Risk Analysis for a Telecoms Provider

Dataset: IBM Telco Customer Churn- 7,043 customer records across demographics, billing, contract type, and service usage

Tools: Python (pandas, scikit-learn, matplotlib, seaborn), SQL (MySQL, DuckDB), Power BI (DAX), GitHub

Deliverables: SQL database and query suite, Python analysis notebook, multi-page Power BI dashboard, project documentation

Key Outcomes

- 7,043 customer records analysed across 21 variables
- 1,869 customers churned : 26.54% churn rate
- £1,669,570.20 in annual revenue lost to churn identified
- £5,487.95 in active revenue at risk surfaced
- Top 50 highest-risk active customers identified for proactive outreach
- A 10% reduction in churn through targeted retention activity would recover an estimated £167,000 annually

1. Data Validation and Quality Assurance

Before drawing any conclusions, the dataset was validated as part of the full data analytics cycle. This step was completed first to ensure no downstream analysis was distorted by data quality issues.

- Ran null checks across all key fields: identified 11 blank TotalCharges records
- Traced the root cause: these were new customers with zero tenure and no billing history, not data entry errors
- Resolved the gap without distorting revenue figures by treating these records appropriately rather than excluding or imputing incorrectly
- Performed duplicate detection and invalid value flagging across demographic, billing, and service fields
- Documented all findings and resolution steps to maintain a clear audit trail before proceeding to analysis

2. SQL Database and Query Design

The dataset was loaded into a relational SQL database (MySQL) to enable structured querying and to demonstrate understanding of relational data concepts required at enterprise scale.

- Designed and built the database schema, mapping customer, contract, billing, and service usage data into a queryable structure
- Wrote targeted queries to calculate churn rates by contract type, payment method, tenure band, internet service type, and number of services
- Built a composite risk-scoring SQL query combining contract type, tenure, monthly charges, and service profile to rank active customers by churn probability
- This query formed the foundation of the Retention Targets page in the Power BI dashboard, surfacing the 50 highest-risk active customers for proactive outreach
- Also used DuckDB for rapid in-notebook querying during exploratory analysis, demonstrating flexibility across SQL environments

3. Exploratory Analysis and Churn Driver Identification

With the dataset validated, analysis was conducted across all key customer dimensions to identify which factors most strongly predicted churn.

- Contract type: Month-to-month customers churn at 42.71% vs 11% for one-year and 3% for two-year contracts. Month-to-month customers account for 86.9% of all revenue lost to churn despite being one segment of the base
- Tenure: Customers in their first 0-6 months churn at 52.94%, the highest of any tenure band. Churn falls consistently with each subsequent band (36%, 29%, 20%, 10%), identifying onboarding as the highest-impact intervention window
- Payment method: Electronic check customers churn at 45%, more than double the rate of automatic payment customers (credit card 15%, bank transfer 17%), pointing to a correlation between payment friction and churn risk
- Internet service: Fibre optic customers churn at 42% vs 19% for DSL and 7% for no internet service, a finding that suggests service quality or pricing issues in the fibre product
- Paperless billing: Customers on paperless billing churn at 34% vs 16% for paper billing
- Number of services: Churn falls sharply as customers subscribe to more services from 46% for single-service customers to 5% for customers with six services confirming that product bundling is a retention lever

4. Revenue Impact Analysis

Churn was translated into financial terms to give stakeholders a clear picture of the business cost, not just the statistical rate.

- Calculated monthly and annualised revenue lost by contract type: month-to-month customers account for £120,850/month in lost revenue
- Identified that customers in the 0-6 month tenure band paying above-average monthly charges represent the highest combined financial risk, high probability of leaving AND high individual revenue value
- Quantified total annual revenue lost to churn at £1,669,570.20, providing the financial case for investment in retention activity
- A 10% reduction in churn through targeted retention activity would recover an estimated £167,000 annually, achievable through contract migration and structured onboarding alone

5. Predictive Modelling: Random Forest Churn Probability

A supervised machine learning model was built to move the business from reactive to predictive retention.

- Built a Random Forest classifier using scikit-learn, trained on demographic, billing, contract, and service usage features
- Evaluated model performance using precision, recall, and F1 score to confirm the model was reliable before scoring active customers
- Applied the trained model to score all active (non-churned) customers, generating an individual churn probability score for each
- Combined the model output with the SQL composite risk score to identify and rank the 50 highest-risk active customers
- These 50 customers were surfaced in the Retention Targets page of the Power BI dashboard, enabling the business to prioritise outreach based on evidence rather than intuition

6. Power BI Dashboard

A multi-page Power BI dashboard was produced so that non-technical stakeholders could act on findings without engaging with raw data.

- **Page 1 : Overview:** High-level KPIs (total customers, churn rate, revenue lost), executive summary, churn by contract type, total customers by churn status
- **Page 2 : Churn Drivers:** Churn rate by payment method, tenure band, internet service type, paperless billing, and number of services
- **Page 3 : Revenue Impact:** Scatter of tenure vs monthly charges by churn status, monthly revenue lost and retained by contract type, financial narrative
- **Page 4 : Retention Targets:** Filterable table of highest-risk active customers ranked by churn probability, with contract, tenure, internet service, and monthly charges
- All pages include slicers for Internet Service, Senior Citizen status, and Contract type, enabling filtered views for specific retention campaigns
- DAX measures used to calculate churn rates, revenue figures, and risk scores dynamically within the dashboard
- Business commentary written on each page so findings were accessible to non-technical audiences

7. Targeted Recommendations

Findings were translated into specific, actionable business recommendations with clear rationale.

- **Contract migration:** Target month-to-month customers, particularly those in months 1-6 with incentives to switch to annual contracts. This single action addresses 86.9% of revenue loss exposure
- **Onboarding intervention:** Build a structured onboarding programme for customers in their first six months. The 52.94% churn rate in this window represents the highest-return intervention opportunity in the dataset
- **Payment method migration:** Actively migrate electronic check customers to automatic payment. The 28-30 percentage point churn rate difference is too large to ignore and suggests this is a solvable friction point
- **Fibre optic product review:** Investigate why fibre optic customers churn at more than double the DSL rate. The data points to a product quality, pricing, or expectation-setting issue
- **Bundling and cross-sell:** Use the single-service vs six-service churn rate differential (46% to 5%) to build a bundling strategy, starting with the highest-churn single-service segments

8. End-to-End Ownership

This project was delivered end-to-end as a solo deliverable, taking full accountability for every component of the solution.

- **SQL:** Database design, schema build, all queries including the composite risk-scoring query
- **Python:** Data validation, exploratory analysis, visualisations (matplotlib, seaborn), Random Forest model build and evaluation
- **Power BI:** Four-page dashboard with DAX measures, slicers, and business commentary
- **Documentation:** Consulting-style project summary written for a non-technical audience
- **Version control:** Full project published on GitHub with version history